$= (\text{kg m}^2\ \text{s}^{-2})(\text{m}^{-2}\ \text{K}^{-1}) = \text{kg s}^{-2}\ \text{K}^{-1}$

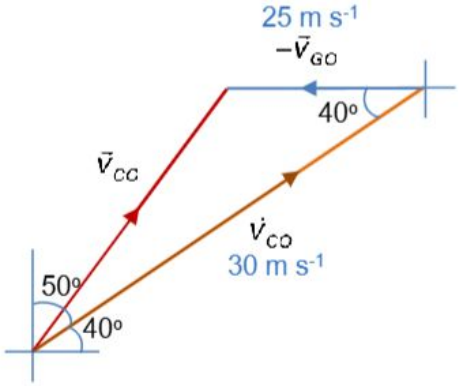
2 **B** Let

- $\vec{v}_{CO}$: velocity of the cheetah
- $\vec{v}_{GO}$: velocity of the gazelle
- $\vec{v}_{CG}$: relative velocity of the cheetah wrt. to the gazelle

$$\vec{v}_{CG} = \vec{v}_{CO} - \vec{v}_{GO}$$

Hence, $v_{CG} = \sqrt{(25^2) + (30^2) - 2(25)(30)\cos(40°)}$

$= 19\ \text{m s}^{-1}$



(Diagram not to scale)

3 **C** Area under the acceleration-time graph is *change* in velocity, hence greatest velocity at 2T.